LECTURE 4:

QUANTITATIVE MODEL FORMULATION: I

OUTLINE

4.1 From Qualitative to Quantitative

4.2 Finite Difference Equations and Differential Equations

4.3 Biological Feedback in Quantitative Models

4.4 Example Model

4.1 From Qualitative to Quantitative

- The boxes of Forrester diagrams represent the objects of interest: the variables whose dynamic quantities we wish to determine over time.
- For the boxes of Forrester diagrams, we must supply a state (dynamic) equation that relates the value of the variable at the next point in the future with the current value and all of the inputs to and outputs from the variable's box.
- The rate equations involve the parameters, auxiliary equations, and driving variables as specified by the Forrester diagram.

4.2 Finite Difference Equations and Differential Equations

- Finite Difference Equations
  - The finite difference equations have the general form:

$$N_{t+1} = f(N_t) \tag{4.1}$$

  - The function *f*() can be arbitrarily complicated. For some *f*(), we can isolate N_t as a separate element:

$$N_{t+1} = N_t + f(\text{state variables, parameters}, t) \tag{4.2}$$

  - Suppose *f*() = *rN*, which is the classical ecological model for density-independent population growth:

$$N_{t+1} = N_t + rN_t \tag{4.3}$$

4.2 Finite Difference Equations and Differential Equations

- Finite Difference Equations

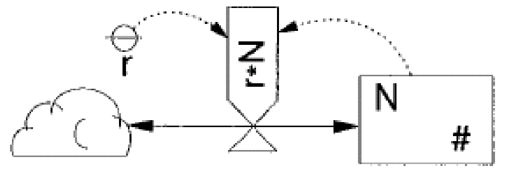


Figure 4.1: Forrester diagram for density-independent population growth.

  - Classical analytical solution to the density-independent growth model in discrete time:

$$
\begin{aligned}
N_1 &= N_0 + rN_0 = N_0(1+r) \\
N_2 &= N_1 + rN_1 = N_1(1+r) = N_0(1+r)(1+r) = N_0(1+r)^2 \\
N_3 &= N_2 + rN_2 = N_0(1+r)^3 \\
&\vdots \\
N_{t+1} &= N_t(1+r) = N_0(1+r)^{t+1}
\end{aligned}
\tag{4.4}
$$

4.2 Finite Difference Equations and Differential Equations

- Finite Difference Equations
  - Recursive finite difference equations assume time is discrete. This implies that no events or processes occur between increments of time.
  - Many biological systems match this situation to a satisfactory degree. An example is the life cycle of an insect that breeds synchronously in the fall. Birth and death in this case defines the discrete nature of time.
  - When we use finite difference equations we are asserting that time and biological processes are discontinuous and that the equations are exact representations

4.2 Finite Difference Equations and Differential Equations

- Differential Equations
  - Differential equations are the continuous time version of finite difference equations.
  - The derivative of a function y with respect to a single variable x is:

$$\frac{dy}{dx} = \lim_{\Delta x \to 0} \frac{y_{x+\Delta x} - y_x}{\Delta x} \tag{4.5}$$

  - Two general approaches to obtaining the integral:
    - Treating the integral as a summation.
    - Applying the rules of integration.

4.2 Finite Difference Equations and Differential Equations

Differential Equations

- Treating the integral as a summation:

$$y_{i+\Delta x} \cong y_i + \underbrace{(2x_i)}_{derivative} \Delta x \qquad (4.6)$$

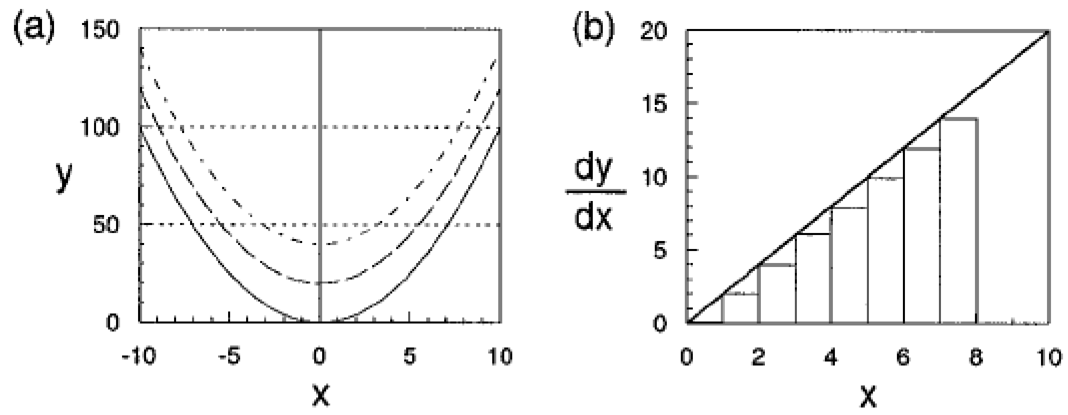


Figure 4.2: (a) The parabola $y = x^2 + C$, where C is an arbitrary constant. (b) The derivative of $y = x^2 + C$, $dy/dx = 2x$ (solid line) and a discretization of the derivative.

4.2 Finite Difference Equations and Differential Equations

- Differential Equations
  - Applying the rules of integration:

$$\frac{dy}{dx} = 2x$$

$$dy = 2xdx \qquad \leftarrow \text{separate variables}$$

$$\int dy = \int 2xdx$$

$$\int dy = y + C_1 \qquad \leftarrow \text{integrate left side}$$

$$\int 2xdx = x^2 + C_2 \qquad \leftarrow \text{integrate right side}$$

Equating these integrals gives

$$y = x^2 + C$$

4.2 Finite Difference Equations and Differential Equations

- Integrating ODEs
  - An ordinary differential equation (ODE) is any equation involving a derivative of a dependent variable with respect to its independent variable.
  - The ODE of the density-independent population model in ecology:

$$\frac{dN}{dt} = rN \qquad (4.7)$$

  - There are, as before, two general strategies for finding the integral: apply the rules of integration, or approximate the area under a curve by summing.

4.2 Finite Difference Equations and Differential Equations

- Integrating ODEs
  - Applying the rules of integration:

$$\frac{dN}{dt} = rN$$

$$\frac{dN}{N} = rdt \qquad \leftarrow \text{separate variables}$$

$$\int \frac{1}{N} dN = \ln N + C_1 \qquad \leftarrow \text{integrate left side}$$

$$\int rdt = r\int dt = rt + C_2 \quad \leftarrow \text{integrate right side}$$

$$\ln\ N = rt + C_3$$

$$N_t = e^{rt+C_3} = e^{C_3} e^{rt} = N_0 e^{rt}$$

  - Summation technique using *Euler approximation*

$$N_{t+\Delta t} \cong N_t + \underbrace{(rN_t)}_{derivative} \Delta t \qquad (4.8)$$

4.3 Biological Feedback in Quantitative Models

From Forrester Diagram to Equations

1. **The first rule** is that every level in a Forrester diagram is a state variable that requires a differential (or difference) equation.
2. **The second rule** is that, at a minimum, every material flow into and out of a state variable requires an explicit algebraic expression. The sum of these expressions associated with the inflow and outflow arrows is the right-hand side of the differential equation.

$$\frac{dx}{dt} = \sum \text{inflows} - \sum \text{outflows}$$

3. **The third rule** is that although biological systems are complex, many of them share a few basic processes that have similar mathematical expressions.

4.3 Biological Feedback in Quantitative Models

- Constant and Bulk Flow Rates

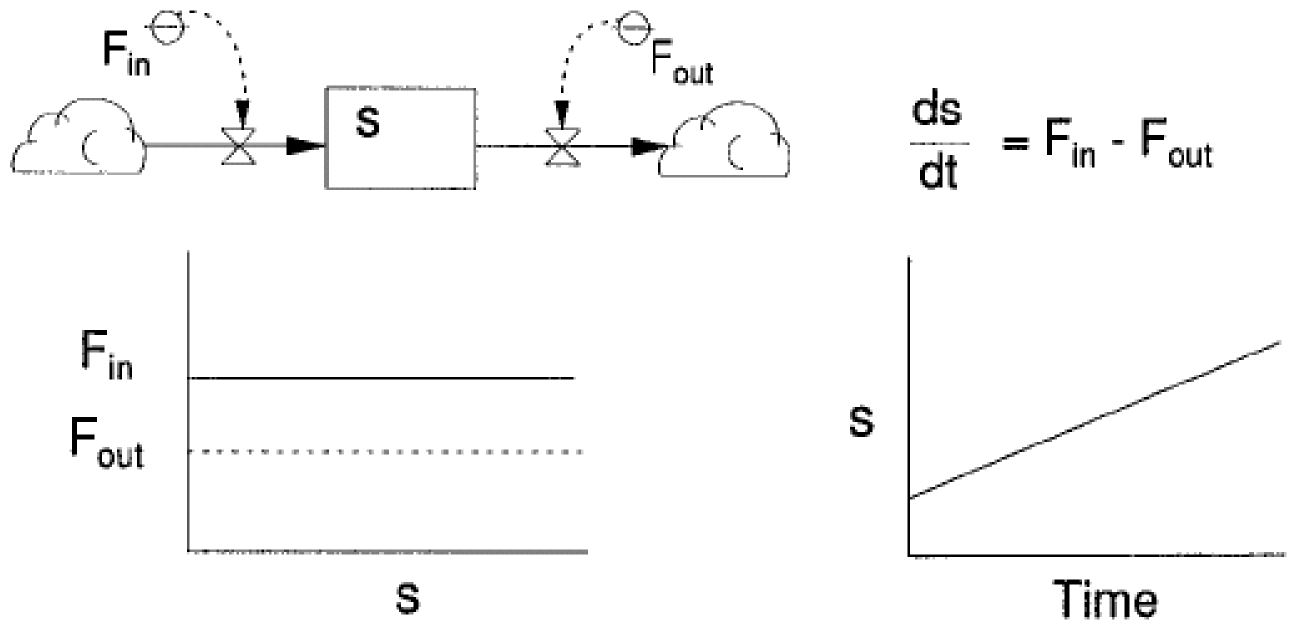


Figure 4.3: Constant rate of flow into a state variable.

4.3 Biological Feedback in Quantitative Models

Constant and Bulk Flow Rates

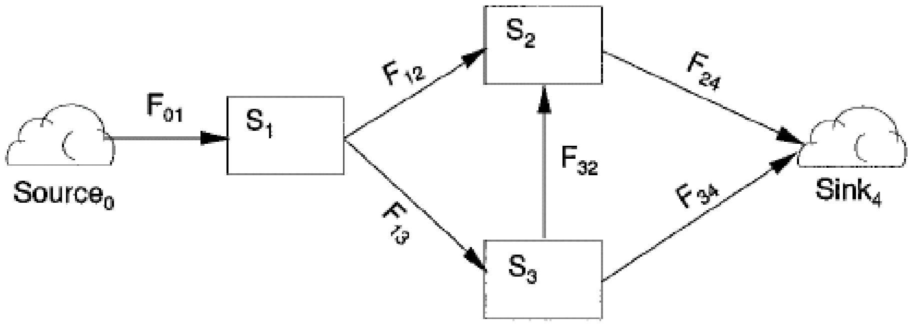


Figure 4.4: Modified Forrester diagram for constant rates of flow among three state variables.

$$
\begin{aligned}
\frac{dS_1}{dt} &= F_{01} - F_{12} - F_{13} \\
\frac{dS_2}{dt} &= F_{12} + F_{32} - F_{24} \\
\frac{dS_3}{dt} &= F_{13} - F_{32} - F_{34}
\end{aligned}
\qquad (4.9)
$$

4.3 Biological Feedback in Quantitative Models

Dynamic Relative Rates

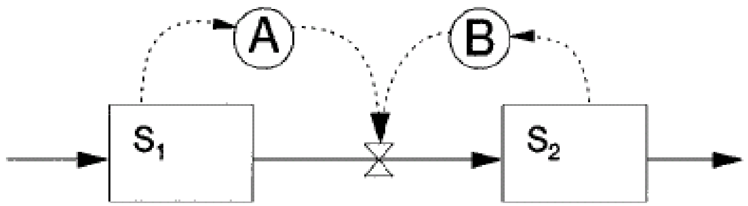


Figure 4.5: Simple information transfer illustrating the influences of state variables on rates.

$$\frac{dS_1}{dt} = \cdots - (A)S_1 - \cdots \tag{4.10}$$

$$\frac{dS_1}{dt} = \cdots - (B)S_2 - \cdots \tag{4.11}$$

$$\frac{dS_1}{dt} = \cdots - (A)S_1(B)S_2 - \cdots \tag{4.12}$$

4.3 Biological Feedback in Quantitative Models

Dynamic Relative Rates

- The quantities A and B are *relative or per capita rates*.
- An example of the island biogeography model:

$$\frac{dR}{dt} = I_x - (I_x / P)R - (E_x / P)R = I_x - ((I_x + E_x)/P)R \qquad (4.13)$$

- Using per capita rates, parameters estimated on individuals can be scaled to the population if we assume all individuals are identical.
- When influence B is absent (in Fig. 4.5 and Eq. 4.10), we say the flow is *donor controlled*, since the "donating" variable (S_1) determines the rate. When A is absent (Eq. 4.11), the flow is *recipient controlled*.

4.3 Biological Feedback in Quantitative Models

Feedback

- *Feedback* is pervasive in biological systems and is one of the fundamental processes that is contained in almost all interesting models.
- Feedback refers to the relationship in which increases or decreases of the value of one or more *controlling variables* affect the *rate* at which a process occurs.
- The action on the rate can be direct or indirect and either positive or negative.
- Any given relation can be either strongly or weakly negative or positive. The balance between the two produces the possibility of sustained oscillations (i.e., dynamics that neither blow up nor return to an equilibrium).

4.3 Biological Feedback in Quantitative Models

- Feedback

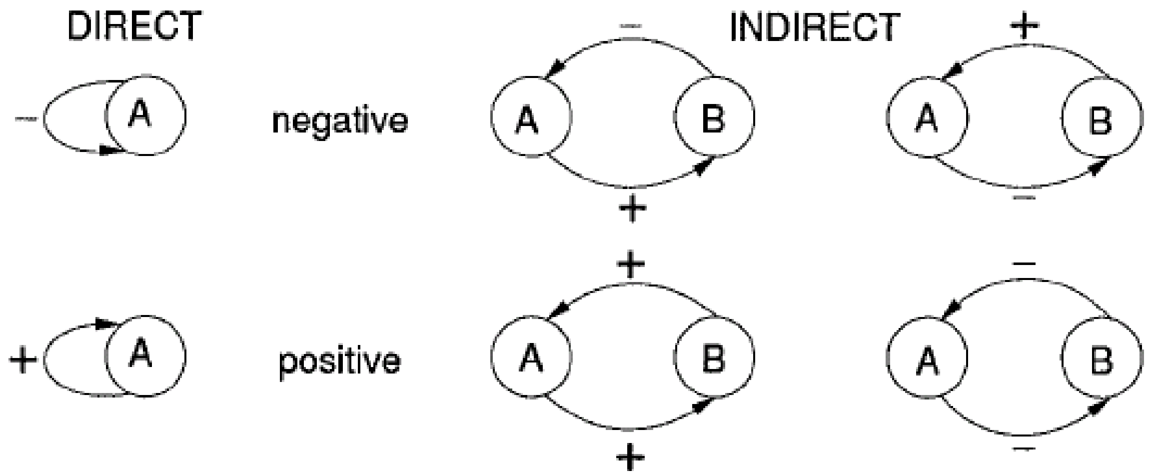


Figure 4.6: Qualitative analysis of direct and indirect effects of system influences producing either positive or negative feedback. The sign on each arc represents the effect of the influencing variable on the variable that terminates the arc.

4.3 Biological Feedback in Quantitative Models

Positive Feedback

- The critical feature of positive feedback is that the rate increases without bound.

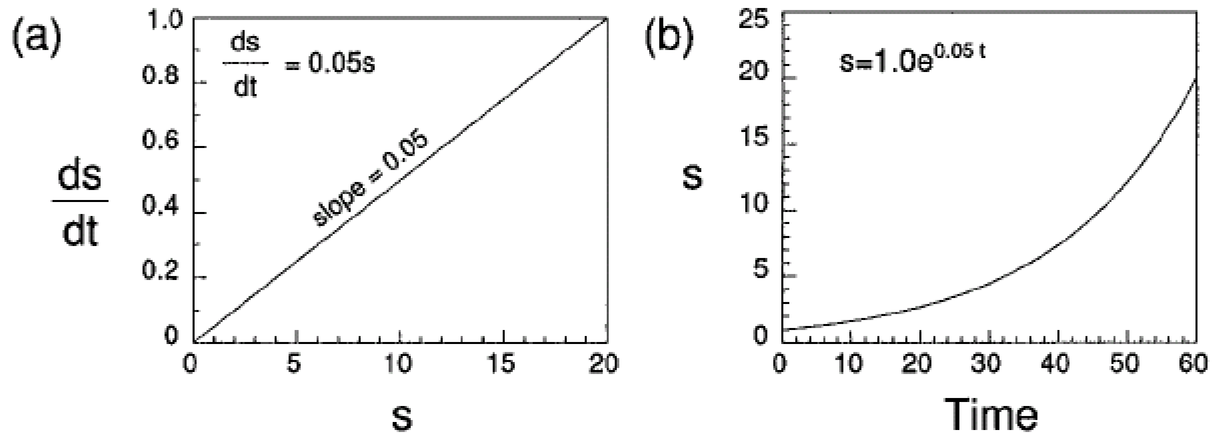


Figure 4.7: Direct positive feedback. (a) Relation of absolute rate of change in a state variable to the value of the variable and a differential equation that behaves in this way. (b) The resulting dynamics.

4.3 Biological Feedback in Quantitative Models

- Negative Feedback
  - The rate of the process is bounded for positive values of the controlling variable.
  - There are three primary mathematical methods by which this condition can be implemented: *feedback by self-inhibition, limitation by extrinsic factors,* and *process saturation*.

4.3 Biological Feedback in Quantitative Models

Self-Inhibition

- The density-dependent, logistic population growth:

$$\frac{dN}{dt} = \left[r\left(1 - \frac{N}{K}\right) \right] N = rN - \frac{r}{K}N^2 \qquad (4.14)$$

$$\frac{dN}{dt}\frac{1}{N} = r - \frac{rN}{K} \qquad (4.15)$$

where *r* is the *maximum per capita rate of change*, and *K* is the *carrying capacity* of the population. Since this model has a single state variable, *N* is *the controlling state variable*.

4.3 Biological Feedback in Quantitative Models

- Self-Inhibition
  - The density-dependent, *logistic* population growth:

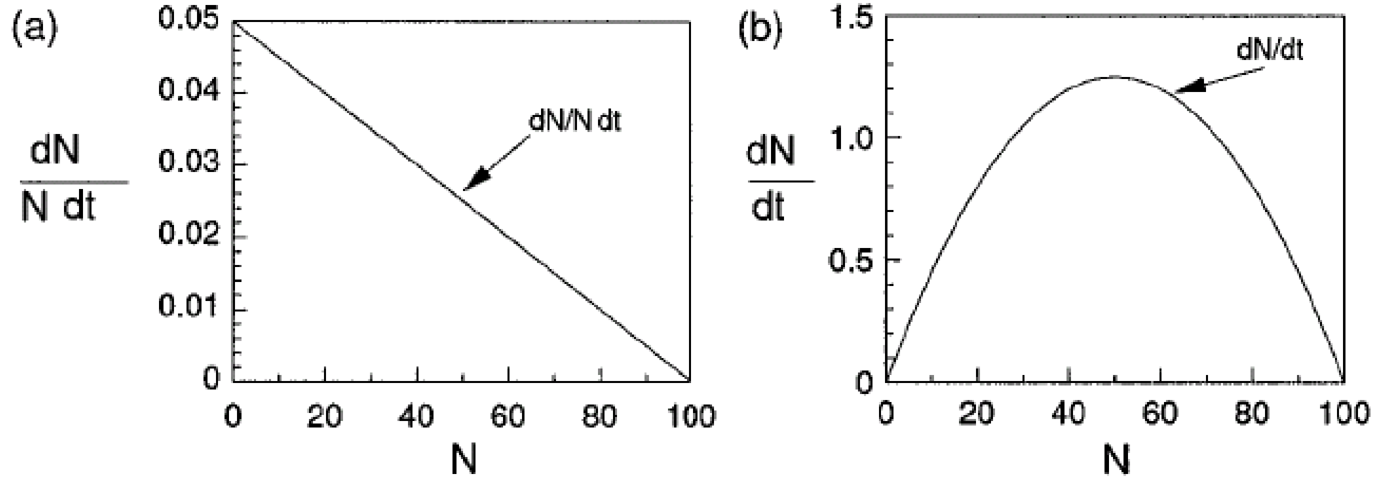


Figure 4.8: (a) Per capita rate of change in density-dependent model as a function of population size. (b) Absolute rate of change in density-dependent model as a function of population size.

4.3 Biological Feedback in Quantitative Models

- Ratios
  - An example of additive self-inhibition:

$$\frac{dy}{dt} = ay - cy^2$$

  - An example of multiplicative self-inhibition (for y > 1):

$$\frac{dy}{dt} = b / y$$

  - A safer formulation of the above equation, which limits *dy/dt* to b as y approaches 0:

$$\frac{dy}{dt} = b\frac{1}{1+y}$$

4.3 Biological Feedback in Quantitative Models

- Extrinsic
  - The Newton's Law of Cooling:

$$\frac{dT}{dt} = k(T_a - T)$$

  - By an extrinsic factor, we mean any quantity "outside" of the state variable to which the differential equation applies. This other quantity may be in the nebulous "unmodeled" environment (e.g., ambient temperature) or it may be the current state or associated auxiliary variable of another modeled state variable.
  - Extrinsic factors are particularly important when we model a flow of materials or energy over a physical distance.

4.3 Biological Feedback in Quantitative Models

- Saturation
  - Negative feedback frequently emerges in systems through an interaction between the quantity of the donor variable and the ability of the recipient to convert the donor substance.
  - By analogy with chemical dynamics where this is common, negative feedback puts bounds on rates by *saturating* the recipient.
  - Saturation is a case where the relation has elements of both positive and negative feedback: the rate neither decreases to 0 nor does it increase indefinitely.
  - The overall dynamical effect is feedback intermediate between positive and negative which permits persistent *oscillations* to occur.

4.3 Biological Feedback in Quantitative Models

- Saturation
  - The Michaelis-Menten model of enzyme kinetics:

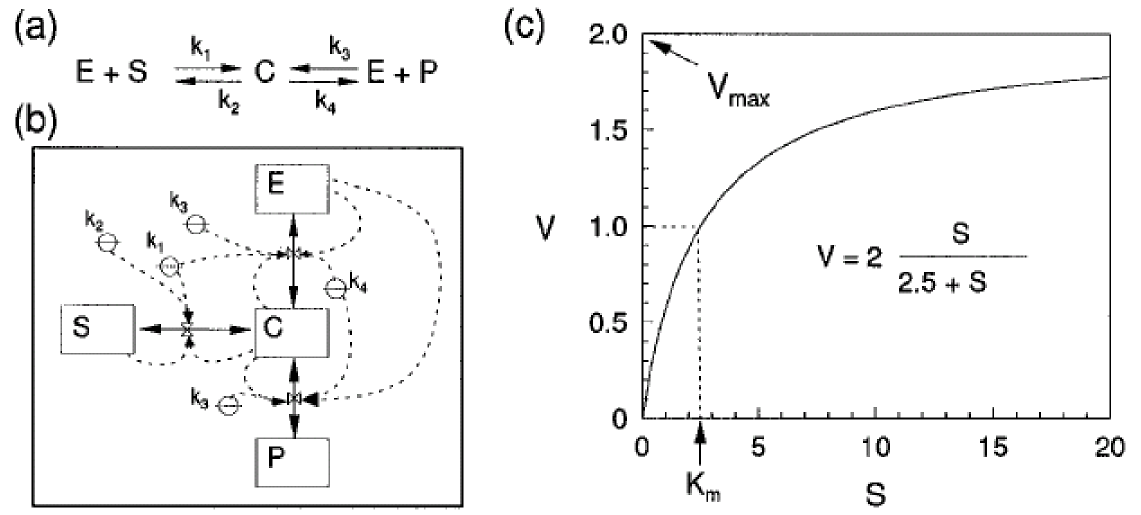


Figure 4.9: Michaelis–Menten saturation feedback control of chemical dynamics: (a) the chemical diagram, (b) the Forrester diagram assuming conserved units, and (c) the rate of formation (V) of the product (P). E represents enzyme concentration, S is the concentration of the substrate, C is the complex formed by the chemical binding of E and S, and P is the product. k_i are the rates of conversion.

4.3 Biological Feedback in Quantitative Models

Saturation

- The Michaelis-Menten model of enzyme kinetics:

$$\frac{dE}{dt} = -k_1 ES + k_2 C + k_4 C - k_3 EP \qquad (4.16)$$

$$\frac{dS}{dt} = -k_1 ES + k_2 C \qquad (4.17)$$

$$\frac{dC}{dt} = k_1 ES - k_2 C - k_4 C + k_3 EP \qquad (4.18)$$

$$\frac{dP}{dt} = k_4 C - k_3 EP \qquad (4.19)$$

- Assuming that (a) the experiments are performed when P is present only at negligible concentrations, and (b) the rate of formation of C equals its breakdown rate, the rate of P formation is described by the Michaelis-Menten equation:

$$V = V_{\max}\left[\frac{S}{K_m + S}\right] \qquad (4.20)$$

4.3 Biological Feedback in Quantitative Models

- Saturation
  - This basic curve is scaled (parameterized) by two parameters: V_{max} (the maximum reaction velocity) scales the velocity to which the curve is asymptotic at large S ; K_m scales how "fast" the curve rises toward the asymptote.
  - The *Holling disc equation* which relates the numbers of prey (y) consumed by a predator in a fixed period of time (e.g., 1 day or 1 experiment duration) to the density of the prey available:

$$y = aT_T\left[\frac{x}{1+ahx}\right] \qquad (4.21)$$

4.3 Biological Feedback in Quantitative Models

Combined Feedback Interaction

- In some systems, saturation or positive feedback can combine with inhibition to produce more complicated relations between variables and rates.
- Usually, this general phenomenon of combined feedback is produced by the action of two or more biological mechanisms (e.g., light saturation of photoreceptors and degradation of enzyme systems at high light intensities, or mate location and competition). Consequently, this situation is frequently modeled as the product of two separate factors.
- For example, photoinhibition can be modeled as:

$$P = P_{\max}\left[\underbrace{(aI)}_{increase}\underbrace{(e^{1-aI})}_{decrease}\right] \qquad (4.22)$$

4.3 Biological Feedback in Quantitative Models

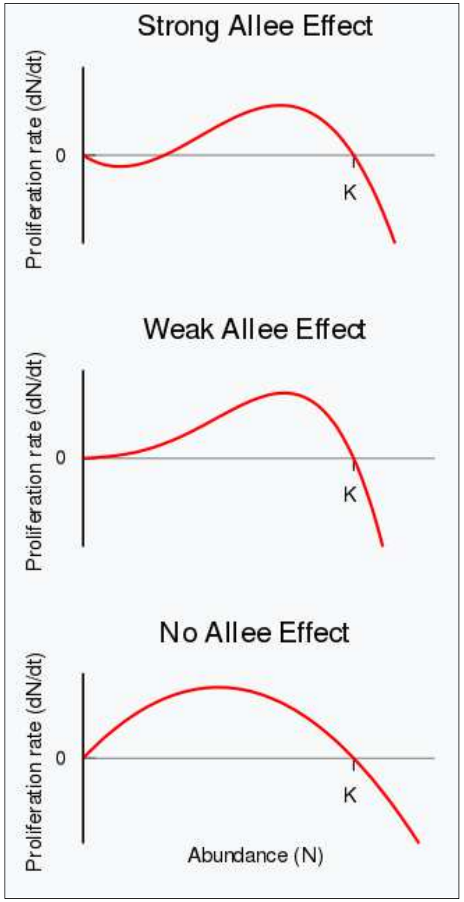
- Saturation
  - Allee effect: The Allee effect is a phenomenon in biology characterized by a positive correlation between population density and the per capita population growth rate in very small populations.
  - A simple example of an Allee effect given by the cubic growth model

$$\frac{dN}{dt} = rN\left(\frac{N}{A}-1\right)\left(1-\frac{N}{K}\right)$$

where the population has a negetive growth rate for 0 < N < A, and a positive growth rate for A < N < K (assuming 0 < A < K). A is called the Allee threshold.

4.3 Biological Feedback in Quantitative Models

- Saturation
  - Allee effect



4.3 Biological Feedback in Quantitative Models

Mass Action

- The Law of Mass Action states that the rate of a reaction is proportional to an integral power of the concentrations of all substances taking part in the reactions.

$$R = aQ^{\alpha}P^{\beta}$$

  where *a* is a constant of proportionality, and α and β are integer powers. The *order* of the reaction relative to *Q* or *P* is α and β, respectively. The order of the overall reaction is the sum of the powers.

- The values of the orders of the relations are often determined by the *stoichiometric* or weight relations of the compounds involved in the reaction.

$$\mathrm{A} + 2\mathrm{B} \underset{k_2}{\overset{k_1}{\rightleftarrows}} \mathrm{C}. \qquad \frac{dC}{dt} = k_1AB^2 - k_2C$$

4.3 Biological Feedback in Quantitative Models

Mass Action

- An example of the classical Lotka-Volterra predator-prey equations:

$$\frac{dV}{dt} = \underbrace{rV}_{\text{positive feedback}} - \underbrace{aVP}_{\text{mass action}} \qquad (4.23)$$

$$\frac{dP}{dt} = \underbrace{abVP}_{\text{conversion}} - \underbrace{dP}_{\text{death}} \qquad (4.24)$$

where the victim (*V*) grows in a density-independent fashion with rate *r*. Predators (*P*) die at a constant per capita rate *d*. The term *aVP* (Eq. 4.23) quantifies the rate at which prey (*V*) are consumed by predators (*P*), so *a* is the search rate. Predators convert the food consumed into new predators with an energetic efficiency *b*.

4.3 Biological Feedback in Quantitative Models

- Multiple Controlling Factors
  - Multiple factors can control a single process rate.
  - There are two different, common situations:
    - The equation for the rate is a univariate function of a primary influencing variable (i.e., the x-axis, such as available light intensity), and one or more of the parameters of this equation is modeled as a function of a second controlling factor (e.g., g C).
    - Second, the rate is the outcome of several interacting factors that combine to create a function having multiple independent variables.

4.3 Biological Feedback in Quantitative Models

- Multiple Controlling Factors
  - Example of the first case:

    A simple model of net photosynthesis rate in plants when it is controlled by both light intensity (I) and carbon availability (C). The primary variable of the rate equation is I and we assume an asymptotic relationship analogous to the Michaelis-Menten relation. The effect of carbon is to increase linearly the maximum rate:

$$P = \frac{\alpha I P_{\max}}{\alpha I + P_{\max}}$$

$$P_{\max} = bC$$

$$P = \frac{\alpha b I C}{\alpha I + bC} \qquad (4.25)$$

4.3 Biological Feedback in Quantitative Models

Multiple Controlling Factors

- The second case concerns multiple factors affecting a process that requires all of the factors.
- The biological case of plant growth in the presence of three nutrients (carbon, nitrogen, and phosphorus).

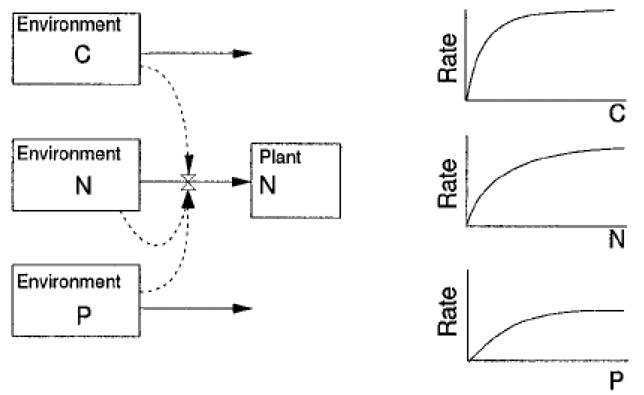


Figure 4.10: Plant growth in which three nutrients interact. On the right is shown Monod growth curves as determined by single-variable experiments that hold the other two nutrients constant.

4.3 Biological Feedback in Quantitative Models

Methods to Combine the Controlling Variables

Consider the simple case with just a single limiting resource (N). Nutrient uptake across cell walls is mediated by ATP and enzymes, so we use Michaelis-Menten kinetics to relate biomass increase to nutrient concentration. When applied to growth rates, we have the *Monod equation*:

$$\frac{dB_N}{dt} = \underbrace{\mu_N^* \left(\frac{N}{K_{m_N} + N} \right)}_{\mu} B_N$$

where μ^*_N is the maximum rate of incorporation of N into plant material per g N of plant material (i.e., a relative or per capita rate). The product of μ^*_{N} and the expression in parentheses is μ (the actual relative rate).

4.3 Biological Feedback in Quantitative Models

- Methods to Combine the Controlling Variables
  - Leibig's Law of the Minimum:

$$\mu = \mu^{*}\left\{\min\left[\left(\frac{C}{K_{m_C}+C}\right),\left(\frac{N}{K_{m_N}+N}\right),\left(\frac{P}{K_{m_P}+P}\right)\right]\right\}$$

  - Multiplicative Rates:

$$\mu = \mu^{*}\left[\left(\frac{C}{K_{m_C}+C}\right)\cdot\left(\frac{N}{K_{m_N}+N}\right)\cdot\left(\frac{P}{K_{m_P}+P}\right)\right]$$

  - Arithmetic Average Rate:

$$\mu = \mu^{*}\frac{1}{3}\left[\left(\frac{C}{K_{m_C}+C}\right)+\left(\frac{N}{K_{m_N}+N}\right)+\left(\frac{P}{K_{m_P}+P}\right)\right]$$

4.3 Biological Feedback in Quantitative Models

- Methods to Combine the Controlling Variables
  - Mean Resistance (Harmonic Mean):

$$S_{eff} = \frac{S}{B+S} \tag{4.26}$$

Using the resistance analogy, the integrated effect (I_{eff}) is computed from:

$$\frac{1}{I_{eff}} = \left(\frac{1}{C_{eff}} + \frac{1}{N_{eff}} + \frac{1}{P_{eff}} \right)$$

$$\frac{1}{I_{eff}} = \left(\sum_{i=1}^{n} \frac{1}{S_{eff,i}} \right)$$

$$H_{eff} = \frac{n}{\left(\sum_{i=1}^{n} \frac{1}{S_{eff,i}} \right)}$$

$$\mu = \mu^* H_{eff}$$

4.3 Biological Feedback in Quantitative Models

- Methods to Combine the Controlling Variables
  - Additive Rates (O'Neill et al., 1989):

$$PI_{eff} = P\frac{CN}{k_2N + CN + k_1C}$$

  - Summary of Multiple Controls:
    - Either replace a constant with a function of the secondary controlling variables (case 1), or use a form of competing factors (case 2).
    - In the second case, the harmonic mean and the Law of the Minimum seem to be the most reasonable forms to use, but this can depend on the system.
    - If the individual functional forms are Michaelis-Menten, then consider using the additive method.

4.4 Example Model

- **The Chemostat Model**
  - A chemostat is a piece of laboratory equipment that grows microbes in a flow-through system of constant volume, *V*, that continuously delivers a constant concentration of nutrients to the population.

$$\frac{dR_1}{dt} = (P/V)(R_{10} - R_1) - N\frac{\mu^*}{Y_1}\min\left[\frac{R_1}{R_1 + K_{m_1}}, \frac{R_2}{R_2 + K_{m_2}}\right]$$

$$\frac{dR_2}{dt} = (P/V)(R_{20} - R_2) - N\frac{\mu^*}{Y_2}\min\left[\frac{R_1}{R_1 + K_{m_1}}, \frac{R_2}{R_2 + K_{m_2}}\right] \qquad (4.27)$$

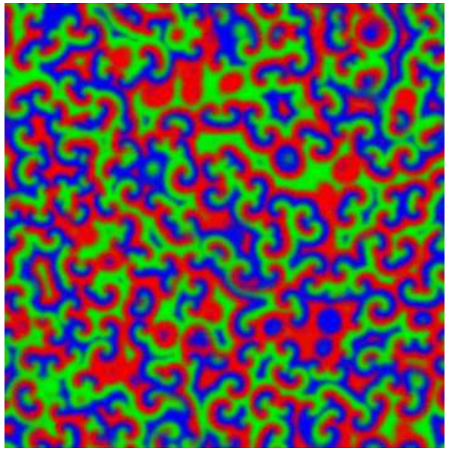
$$\frac{dN}{dt} = N\mu^*\min\left[\frac{R_1}{R_1 + K_{m_1}}, \frac{R_2}{R_2 + K_{m_2}}\right] - (P/V)N$$

where μ^* is max[μ_1, μ_2], Y_i is a constant to convert cell numbers to appropriate nutrient units, and K_{m_i} are the half-saturation constants.

4.4 EXAMPLE MODEL

- The Belousov-Zhabotinsky Reaction Model
  - The Belousov-Zhabotinsky (BZ) reaction is a classical example of a non-equilibrium chemical oscillator in which the components exhibit periodic changes in concentration. The Field and Noyes model (1974) can be expressed as follows.

$$A+Y \rightarrow X$$
$$X+T \rightarrow P$$
$$B+X \rightarrow 2X+Z$$
$$2X \rightarrow Q$$
$$Z \rightarrow fY$$



This relates to the chemical mechanism by $X=HBrO_2$, $Y=Br^-$, $Z=Ce^{4+}$. A, B, P, Q are non-intermediates and can be regarded as constants.

4.4 Example Model

- The Belousov-Zhabotinsky Reaction Model
  - The Field and Noyes model (1974) expressed in differential equations:

$$\frac{dX}{dt} = k_1 AY - k_2 XY + k_3 BX - 2k_4 X^2$$

$$\frac{dY}{dt} = -k_1 AY - k_2 XY + fk_5 Z$$

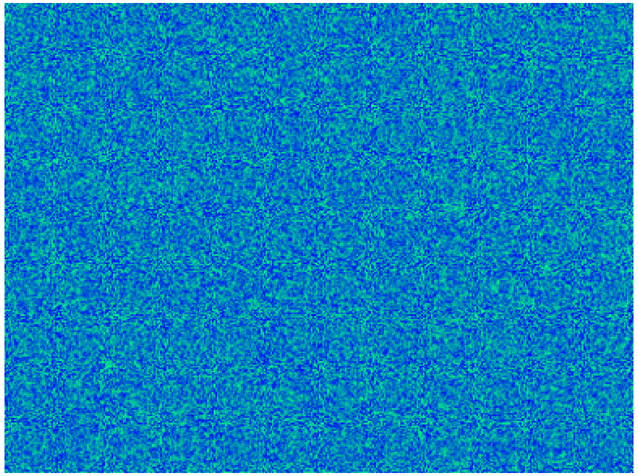
$$\frac{dZ}{dt} = k_3 BX - k_5 Z$$

  - A simpler reaction model by Turner (2009) can be expressed as: (α, β, γ are rate constants)

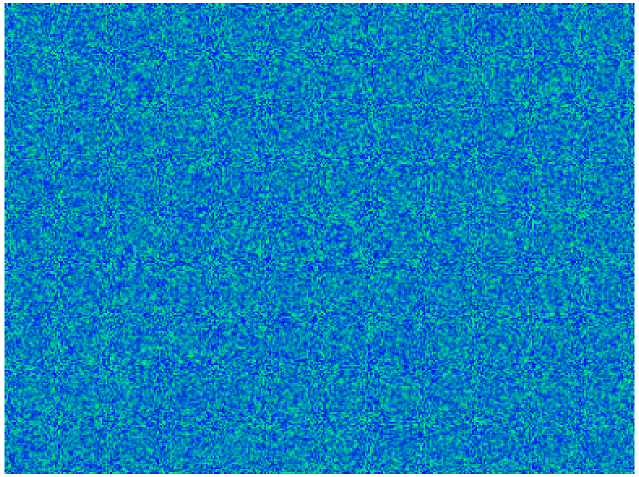
$$\begin{array}{lll} \mathrm{A} + \mathrm{B} \to 2A & \alpha & [\mathrm{A}]_{t+1} = [\mathrm{A}]_t + [\mathrm{A}]_t(\alpha[\mathrm{B}]_t - \gamma[\mathrm{C}]_t) \\ \mathrm{B} + \mathrm{C} \to 2B & \beta & [\mathrm{B}]_{t+1} = [\mathrm{B}]_t + [\mathrm{B}]_t(\beta[\mathrm{C}]_t - \alpha[\mathrm{A}]_t) \\ \mathrm{C} + \mathrm{A} \to 2C & \gamma & [\mathrm{C}]_{t+1} = [\mathrm{C}]_t + [\mathrm{C}]_t(\gamma[\mathrm{A}]_t - \beta[\mathrm{B}]_t) \end{array}$$

4.4 Example Model

- The Belousov-Zhabotinsky Reaction Model
  - Spirals and waves appear spontaneously and unpredictably in the concentration profiles of the reaction components on a scale dictated by the values of α, β and γ.



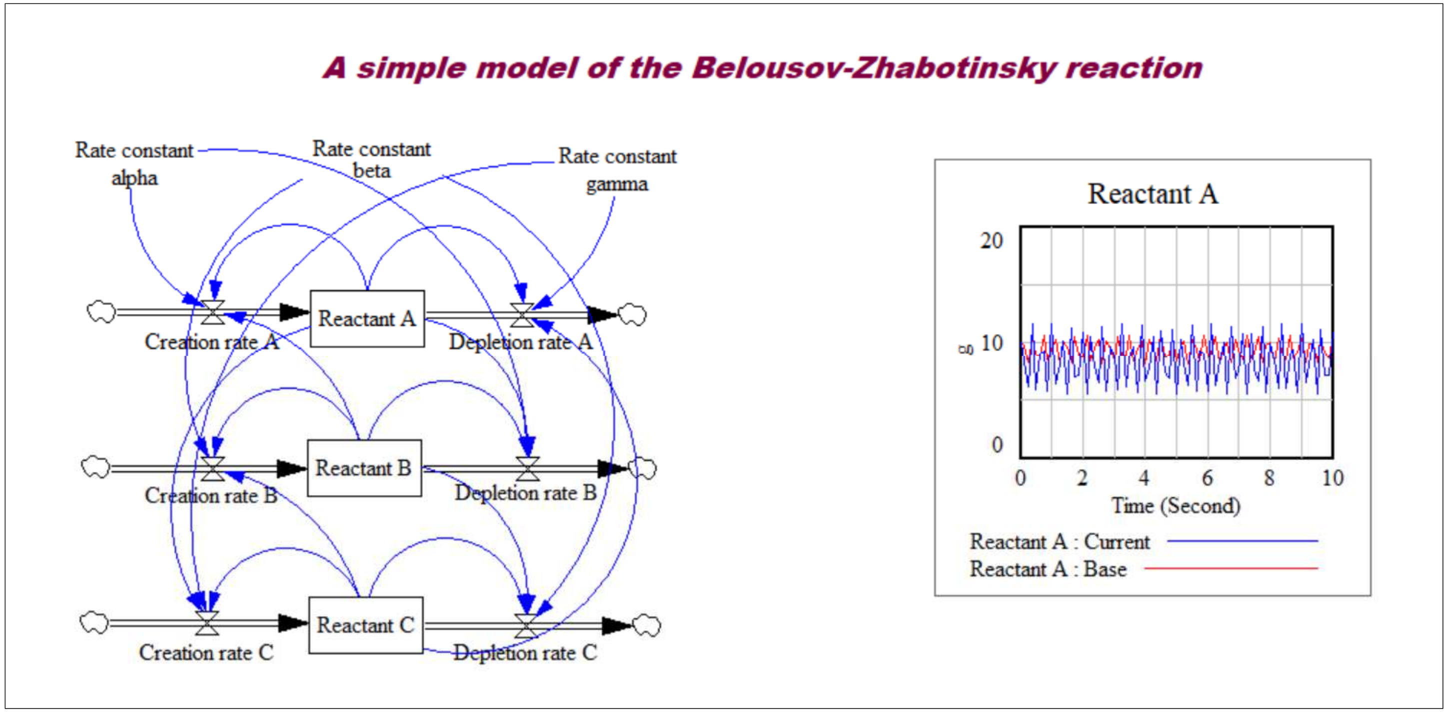
For α=β=γ=1



For α=1.2, β=γ=1

4.4 Example Model

- The Belousov-Zhabotinsky Reaction Model
  - Vensim Implementation



α=1.2, β=γ=1 (Base) compared with α=1.6, β=γ=1 (Current)